

Project report

Examination of gene expression in treated Zebrafish embryos – Round 2

Prepared for:
Akeem Gardner
Canurta Inc.
2425 Matheson Blvd. E, Suite 807
Mississauga, ON L4W 5K4

Feb 4th, 2024

Prepared by: David Konkin

David Konkin
National Research Council Canada
110 Gymnasium Pl. Saskatoon, SK



Overview

The goal of this project was to perform RNA-seq based differential gene expression analysis of Zebrafish embryos in order to characterize the response to potential neuroprotective protective agents in chemical model of neurodegeneration (BMAA). This report covers the addition of additional replicates to the 1st experiment.

Results

RNA-seq library preparation and sequencing

Using the same protocol as the initial experiment, an additional fifteen RNA samples isolated from client supplied material from the same rounds of treatments that, based on tube labels, took place between April 17th and May 9th (Table 1). These samples produced high quality RNA consistent with the first batch (Figure 1). The same standard RNA-seq library preparation workflow was employed to prepare the RNA-seq libraries. The resulting libraries were sequenced to an average depth of 36.2 million read pairs (Table 2). Further, additional sequencing coverage was obtained for the original fifteen samples, bringing the average coverage of the original fifteen samples to 36.3 million read pairs. https://bioinfo.nrc.ca/canurta_zebrafish/BMAA_results_5nov2024/BMAA_raw_reads/

Read mapping and differential gene expression

Raw reads were trimmed and mapped to the Zebrafish reference genome. An average of 36.2 million read pairs were mapped per sample (Table 2.). Pair-wise differential gene expression analysis was completed using DEseq2 for all sample pairs. Gene-level differential expression data for each pair are available in tabular csv format at: https://bioinfo.nrc.ca/canurta_zebrafish/BMAA_results_5nov2024/DGE_comparisons/ Descriptions of each field are listed in Appendix 1. Values are relative to the second sample listed in each filename. Redundant transcript annotations have been collapsed in the files prepended with “updated_”, so that there is only a single entry per gene.

Principal components analysis (Figure 2.) of transcript abundance per sample including the additional samples again revealed a fairly wide dispersion of replicates. A similarity heat map (Figure 3.) also highlighted that there appear to be strong source(s) of technical variation causing unexpected clustering. The increased number of replicates allows the identification of outliers such as BMAA-Pure reps 5, 6 and 1, as well as BMAA-rep3 and BMAA-Eda rep5. These five outliers appear to have belonged to experiments on May8th and 9th. To better account for day to day variation in future experiments, conducting a single repliate of each condition together on a single day is recommended.

Table 1. Input, RNA, and RNA-seq library statistics

Initial samples		RNA						Library	
Internal Designation	Designation	Embryo Collected tube notes	RNA RIN Value		RNA Concentration (ng/ul)	Elution Volume (ul)	Total RNA (ng)	Fragment size (bp)	Library nM
C-1	NoTreatment_rep1_May2024	May9/2024	8.8		546	30	16380	348	195.44
C-2	NoTreatment_rep2_May2024	May9/2024	9.6		398	30	11940	338	239.01
C-3	NoTreatment_rep3_May2024	May7/2024	9.4		614	30	18420	345	265.08
C-4	BMAA_rep1_May2024	May8/2024	9.2		246	30	7380	349	155.69
C-5	BMAA_rep2_May2024	May7/2024	9.6		644	30	19320	355	335.70
C-6	BMAA_rep3_May2024	May8/2024	8.8		620	30	18600	333	285.70
C-7	BMAA+Eda_rep1_May2024	May9/2024	9.7		530	30	15900	340	318.26
C-8	BMAA+Eda_rep2_May2024	May7/2024	9.8		544	30	16320	336	354.83
C-9	BMAA+Eda_rep3_May2024	May8/2024	9.8		480	30	14400	352	300.18
C-10	BMAA+Pure_rep1_May2024	May9/2024	9.8		452	30	13560	353	243.06
C-11	BMAA+Pure_rep2_May2024	May8/2024	9.7		482	30	14460	344	240.53
C-12	BMAA+Pure_rep3_May2024	May7/2024	9.8		540	30	16200	343	284.02
C-13	BMAA+ALS_rep1_May2024	May8/2024	9.8		468	30	14040	338	282.48
C-14	BMAA+ALS_rep2_May2024	May8/2024	9.9		340	30	10200	336	285.28
C-15	BMAA+ALS_rep3_May2024	May9/2024	9.8		628	30	18840	341	360.27
Additional samples									
Internal Designation	Designation	Embryo Collected tube notes	RNA RIN Value (Run 1..partial results)	RNA RIN Value (Run2 Complete)	RNA Concentration (ng/ul)	Elution Volume (ul)	Total RNA (ng)	Fragment size (bp)	Library nM
C-16	NoTreatment_rep4	May7/2024	9.1	9.2	714	30	21420	345	184.15
C-17	NoTreatment_rep5	April17/2024	8.6	8.3	654	30	19620	336	154.74
C-18	NoTreatment_rep6	April26/2024	8.9	8.5	496	30	14880	337	152.27
C-19	BMAA_rep4	May9/2024	9	9.5	532	30	15960	342	107.37
C-20	BMAA_rep5	May7/2024	9.8	9.4	832	30	24960	358	144.43
C-21	BMAA_rep6	May7/2024	9.6	9.3	698	30	20940	338	169.39
C-22	BMAA+Eda_rep4	May8/2024	9.8	9.8	590	30	17700	332	172.30
C-23	BMAA+Eda_rep5	May9/2024	9.7	9.6	574	30	17220	330	227.26
C-24	BMAA+Eda_rep6	May7/2024	9.8	9.6	382	30	11460	351	36.53
C-25	BMAA+Pure_rep4	May7/2024	x	9.6	616	30	18480	351	94.30
C-26	BMAA+Pure_rep5	May8/2024	x	9.5	430	30	12900	361	89.26
C-27	BMAA+Pure_rep6	May9/2024	9	9.7	668	30	20040	333	99.78
C-28	BMAA+ALS_rep4	May7/2024	4.1(k)	9.1	714	30	21420	337	176.55
C-29	BMAA+ALS_rep5	May7/2024	9.6	8.8	578	30	17340	343	151.76
C-30	BMAA+ALS_rep6	May9/2024	5.5(k)	9.7	636	30	19080	335	228.25

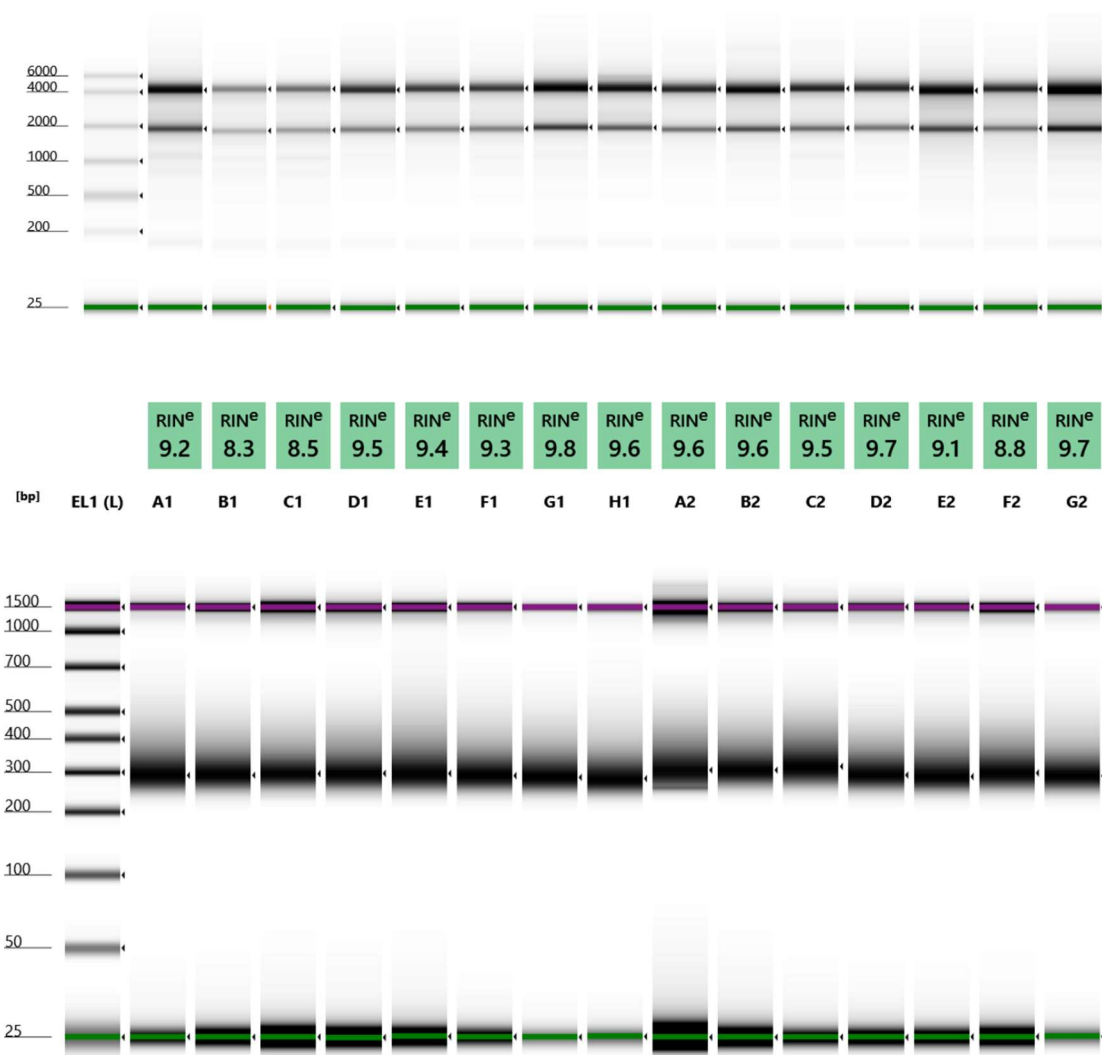


Figure 1. Tapestation-based electrophoretic quality control of input RNA and RNA-seq libraries. Upper pane: Input RNA samples Right pane: Final RNA-seq libraries (C-1 through C-15)

Table 2. RNA-seq read statistics

Sample	file	format	type	num_seqs	sum_len	min_len	avg_len	max_len
C-1	CTRL_rep1_r1.fastq.gz	FASTQ	DNA	42029499	6346454349	151	151	151
C-1	CTRL_rep1_r2.fastq.gz	FASTQ	DNA	42029499	6346454349	151	151	151
C-2	CTRL_rep2_r1.fastq.gz	FASTQ	DNA	38097119	5752664969	151	151	151
C-2	CTRL_rep2_r2.fastq.gz	FASTQ	DNA	38097119	5752664969	151	151	151
C-3	CTRL_rep3_r1.fastq.gz	FASTQ	DNA	34686918	5237724618	151	151	151
C-3	CTRL_rep3_r2.fastq.gz	FASTQ	DNA	34686918	5237724618	151	151	151
C-4	BMAA_rep1_r1.fastq.gz	FASTQ	DNA	42324245	6390960995	151	151	151
C-4	BMAA_rep1_r2.fastq.gz	FASTQ	DNA	42324245	6390960995	151	151	151
C-5	BMAA_rep2_r1.fastq.gz	FASTQ	DNA	42597571	6432233221	151	151	151
C-5	BMAA_rep2_r2.fastq.gz	FASTQ	DNA	42597571	6432233221	151	151	151
C-6	BMAA_rep3_r1.fastq.gz	FASTQ	DNA	34271478	5174993178	151	151	151
C-6	BMAA_rep3_r2.fastq.gz	FASTQ	DNA	34271478	5174993178	151	151	151
C-7	BMAAeda_rep1_r1.fastq.gz	FASTQ	DNA	43928277	6633169827	151	151	151
C-7	BMAAeda_rep1_r2.fastq.gz	FASTQ	DNA	43928277	6633169827	151	151	151
C-8	BMAAeda_rep2_r1.fastq.gz	FASTQ	DNA	35176455	5311644705	151	151	151
C-8	BMAAeda_rep2_r2.fastq.gz	FASTQ	DNA	35176455	5311644705	151	151	151
C-9	BMAAeda_rep3_r1.fastq.gz	FASTQ	DNA	36702125	5542020875	151	151	151
C-9	BMAAeda_rep3_r2.fastq.gz	FASTQ	DNA	36702125	5542020875	151	151	151
C-10	BMAApure_rep1_r1.fastq.gz	FASTQ	DNA	32249159	4869623009	151	151	151
C-10	BMAApure_rep1_r2.fastq.gz	FASTQ	DNA	32249159	4869623009	151	151	151
C-11	BMAApure_rep2_r1.fastq.gz	FASTQ	DNA	30681616	4632924016	151	151	151
C-11	BMAApure_rep2_r2.fastq.gz	FASTQ	DNA	30681616	4632924016	151	151	151
C-12	BMAApure_rep3_r1.fastq.gz	FASTQ	DNA	32958931	4976798581	151	151	151
C-12	BMAApure_rep3_r2.fastq.gz	FASTQ	DNA	32958931	4976798581	151	151	151
C-13	BMAAals_rep1_r1.fastq.gz	FASTQ	DNA	31985677	4829837227	151	151	151
C-13	BMAAals_rep1_r2.fastq.gz	FASTQ	DNA	31985677	4829837227	151	151	151
C-14	BMAAals_rep2_r1.fastq.gz	FASTQ	DNA	29173839	4405249689	151	151	151
C-14	BMAAals_rep2_r2.fastq.gz	FASTQ	DNA	29173839	4405249689	151	151	151
C-15	BMAAals_rep3_r1.fastq.gz	FASTQ	DNA	38010303	5739555753	151	151	151
C-15	BMAAals_rep3_r2.fastq.gz	FASTQ	DNA	38010303	5739555753	151	151	151
C-16	CTRL_rep4_r1.fastq.gz	FASTQ	DNA	39900829	5985124350	150	150	150
C-16	CTRL_rep4_r2.fastq.gz	FASTQ	DNA	39900829	5985124350	150	150	150
C-17	CTRL_rep5_r1.fastq.gz	FASTQ	DNA	33857600	5078640000	150	150	150
C-17	CTRL_rep5_r2.fastq.gz	FASTQ	DNA	33857600	5078640000	150	150	150
C-18	CTRL_rep6_r1.fastq.gz	FASTQ	DNA	38058869	5708830350	150	150	150
C-18	CTRL_rep6_r2.fastq.gz	FASTQ	DNA	38058869	5708830350	150	150	150
C-19	BMAA_rep4_r1.fastq.gz	FASTQ	DNA	35798570	5369785500	150	150	150
C-19	BMAA_rep4_r2.fastq.gz	FASTQ	DNA	35798570	5369785500	150	150	150
C-20	BMAA_rep5_r1.fastq.gz	FASTQ	DNA	43819791	6572968650	150	150	150
C-20	BMAA_rep5_r2.fastq.gz	FASTQ	DNA	43819791	6572968650	150	150	150
C-21	BMAA_rep6_r1.fastq.gz	FASTQ	DNA	39217229	5882584350	150	150	150
C-21	BMAA_rep6_r2.fastq.gz	FASTQ	DNA	39217229	5882584350	150	150	150
C-22	BMAAeda_rep4_r1.fastq.gz	FASTQ	DNA	39492541	5923881150	150	150	150
C-22	BMAAeda_rep4_r2.fastq.gz	FASTQ	DNA	39492541	5923881150	150	150	150
C-23	BMAAeda_rep5_r1.fastq.gz	FASTQ	DNA	44785421	6717813150	150	150	150
C-23	BMAAeda_rep5_r2.fastq.gz	FASTQ	DNA	44785421	6717813150	150	150	150
C-24	BMAAeda_rep6_r1.fastq.gz	FASTQ	DNA	28160371	4224055650	150	150	150
C-24	BMAAeda_rep6_r2.fastq.gz	FASTQ	DNA	28160371	4224055650	150	150	150
C-25	BMAApure_rep4_r1.fastq.gz	FASTQ	DNA	29280464	4392069600	150	150	150
C-25	BMAApure_rep4_r2.fastq.gz	FASTQ	DNA	29280464	4392069600	150	150	150
C-26	BMAApure_rep5_r1.fastq.gz	FASTQ	DNA	30793647	4619047050	150	150	150
C-26	BMAApure_rep5_r2.fastq.gz	FASTQ	DNA	30793647	4619047050	150	150	150
C-27	BMAApure_rep6_r1.fastq.gz	FASTQ	DNA	38132370	5719855500	150	150	150
C-27	BMAApure_rep6_r2.fastq.gz	FASTQ	DNA	38132370	5719855500	150	150	150
C-28	BMAAals_rep4_r1.fastq.gz	FASTQ	DNA	38149904	5722485600	150	150	150
C-28	BMAAals_rep4_r2.fastq.gz	FASTQ	DNA	38149904	5722485600	150	150	150
C-29	BMAAals_rep5_r1.fastq.gz	FASTQ	DNA	31845616	4776842400	150	150	150
C-29	BMAAals_rep5_r2.fastq.gz	FASTQ	DNA	31845616	4776842400	150	150	150
C-30	BMAAals_rep6_r1.fastq.gz	FASTQ	DNA	32175648	4826347200	150	150	150
C-30	BMAAals_rep6_r2.fastq.gz	FASTQ	DNA	32175648	4826347200	150	150	150

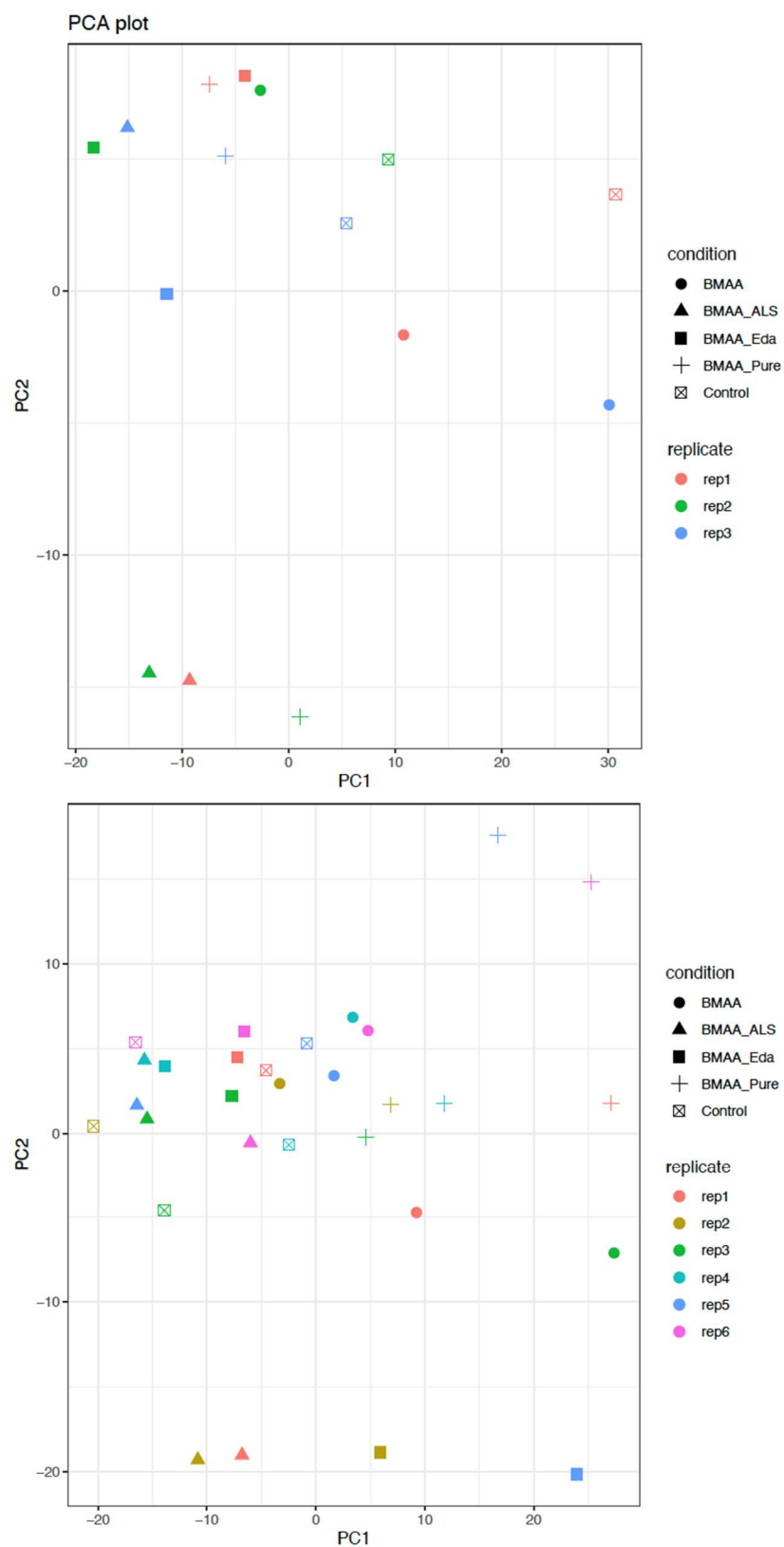


Figure 2. Principal component analysis of transcript abundance for original three replicate experiment (above) and six replicate experiment (below)

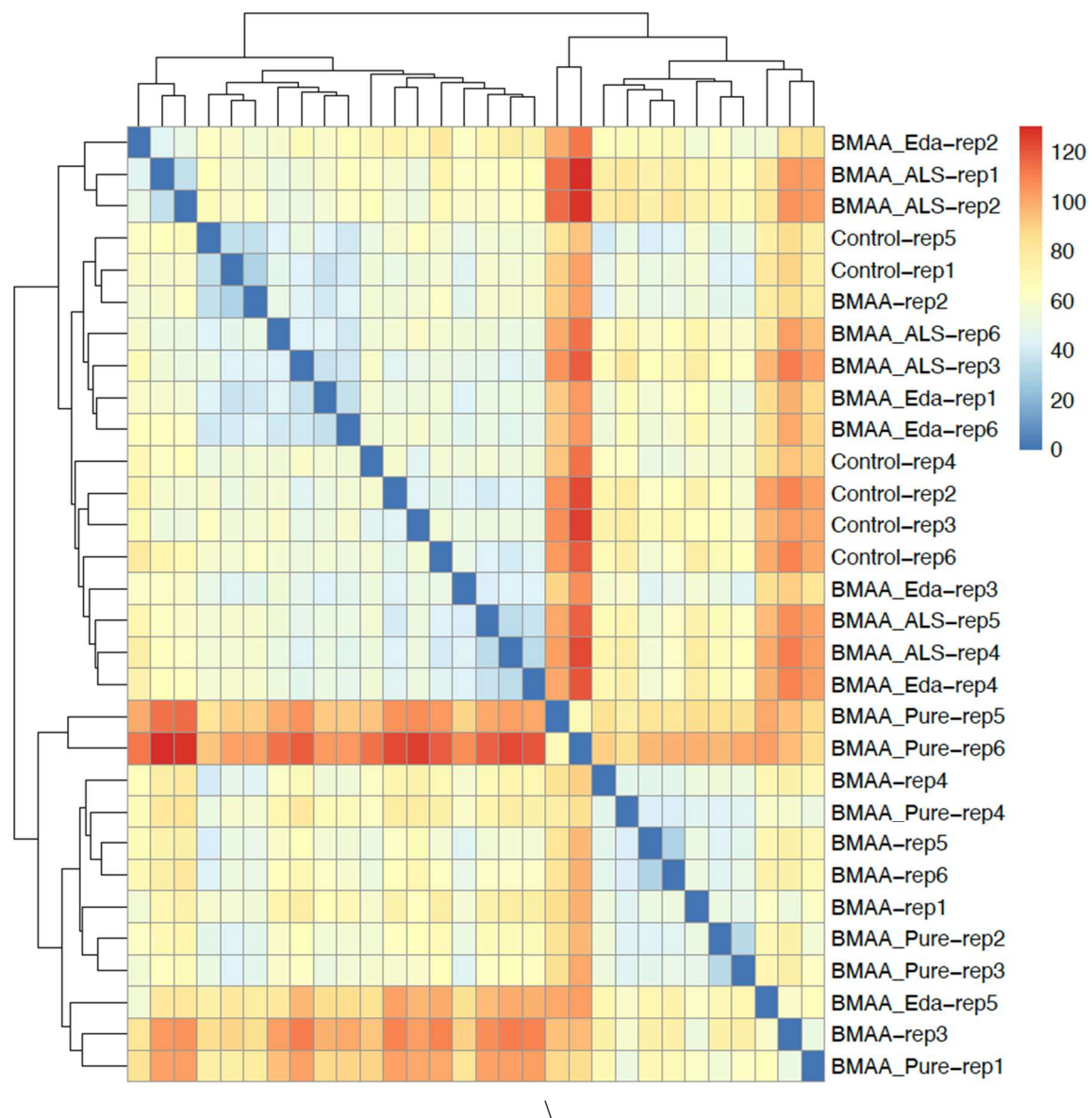


Figure 3. Similarity matrix and clustering of samples by overall gene expression similarity. The heatmap shows degree of similarity between the samples.

With the additional replicates, the number of significantly expressed gene increased tremendously. For example in the BMAA vs control condition, there are 3792 differentially expressed genes compared to 92 for the three replicate analysis. To highlight the impact of the treatments on genes affected by BMAA treatment, we have provided a heatmap of differential expression values, statistical significance and replicate count values for the 3792 differentially expressed genes in the BMAA vs Control and

provided these in the email attachment “canurta_3792_BMAA_DEGs_heatmap.xlsx”. The differential

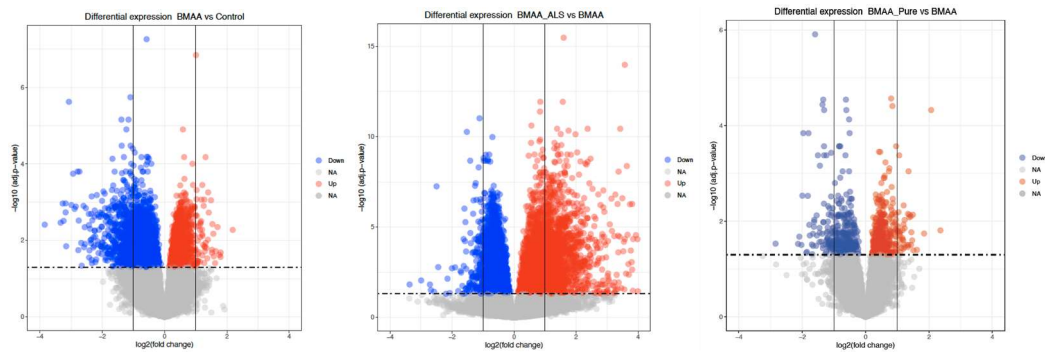


Figure 4. Representative Volcano plots comparing log₂-fold change (X-axis) with adjusted p-value (Y-axis) for BMAA vs Control (left), BMAA+ALS vs BMAA (middle), and BMAA+Pure vs BMAA (right). Vertical solid bar indicate negative and positive 2-fold thresholds. The dotted horizontal bar indicates the adjusted p-value threshold corresponding to 0.05. Datapoints above the adjusted p-value threshold were considered statistically significant and used for downstream GO and KEGG enrichment analyses.

Gene and KEGG functional enrichment analyses

Differentially expressed genes (Table 3) were used to conduct functional enrichment analysis using Gene Ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) metabolic pathway annotations. All figures and tabular results for comparisons that yielded at least one enrichment are provided in:

https://bioinfo.nrc.ca/canurta_zebrafish/BMAA_results_5nov2024/GO_results/

and https://bioinfo.nrc.ca/canurta_zebrafish/BMAA_results_5nov2024/KEGG_results/

These tests for functional enrichment compare the ratio of differentially expressed genes belonging to each functional category to the background ratio.

Conclusions

While the sample to sample variation remains a factor, the inclusion of additional replicates has allowed a much more statistically robust analysis and more clearly documented the response patterns to BMAA and interventions. Further refinement of experimental conditions such as matched replicates across all conditions in a single batch may improve experimental clarity further.

Material and methods

RNA extraction and sequencing library preparation

RNA was extracted using the Qiagen RNeasy extraction kit according to manufacturers instructions. Tissue was homogenized by pipetting the embryos in Buffer RLT plus thirty times. RNA for two samples (C-1 (Control rep1) and C-2 (Control rep 2) were extracted

one day to verify the protocol worked. The remainingsamples were extracted in one batch the following day. RNA-seq libraries were prepared using a the Illumina Truseq Stranded RNA kit according to manufacturer's instruction. Libraries were sequenced on an Illumina Novaseq 6000 using v1.5 2x150 bp chemistry.

Differential gene expression (DEG) analysis

Raw sequencing read files were trimmed using cutadapt (Martin, 2011). Fastqc (Andrews, 2010) was used to analyze the quality of the reads before and after trimming. Reads were aligned and quantified with STAR (Dobin et al., 2013; Dobin and Gingeras, 2015) against the Danio_rerio.GRCz11 in quant mode. Raw read per gene counts from STAR were formatted into a matrix (gene x sample) and imported into DESeq2 (Love et al., 2014) for all differential gene expression analyses and pairwise comparisons. An adjusted p-value (padj, q) false discovery rate (FDR) of 0.05 (Benjamini and Hochberg) was used as the cutoff value for downstream analysis.

Gene ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis

Differentially expressed genes (DEGs) ($q < 0.05$) identified above were used for GO and KEGG analysis using clusterprofiler (Wu et al., 2021). For the GO analysis, the OrgDb used was "org.Dr.eg.db", and for the KEGG analysis the organism used was "dre". Plots and PCA analysis were completed using R.

Appendix 1. Descriptions for table fields

DGE_comparisons

ensembl_gene_id – ensemble gene ID
baseMean – mean of normalized counts for all samples
log2FoldChange - log2 fold change (MLE): condition treated vs untreated
lfcSE – standard error: condition treated vs untreated
stat – Wald statistic: condition treated vs untreated
pvalue – Wald test p-value: condition treated vs untreated
padj – BH adjusted p-values
ensembl_transcript_id_version – ensemble transcript ID version
external_gene_name – gene name
description – gene description
transcript_biotype – transcript biotype

KEGG and GO results table headers

ID – refers to the GO or KEGG group identifier
Description – description of the GO or KEGG group
GeneRatio – genes of interest in the gene set / total genes of interest
BgRatio – genes within each gene set / all unique genes across all gene sets
pvalue – p value statistic
p.adjust – adjusted pvalue
qvalue – qvalue statistic
geneID – list of gene IDs

Count – number of genes in this group

Definitions

For GO enrichment:

BP – biological process

CC – cellular component

MF – molecular function

References

Andrews, Simon. "FastQC: a quality control tool for high throughput sequence data. 2010." (2017).

Dobin, Alexander et al. "STAR: ultrafast universal RNA-seq aligner." *Bioinformatics (Oxford, England)* vol. 29,1 (2013): 15-21. doi:10.1093/bioinformatics/bts635

Dobin, Alexander, and Thomas R Gingeras. "Mapping RNA-seq Reads with STAR." *Current protocols in bioinformatics* vol. 51 11.14.1-11.14.19. 3 Sep. 2015, doi:10.1002/0471250953.bi1114s51

Love, Michael I., Wolfgang Huber, and Simon Anders. "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2." *Genome biology* 15 (2014): 1-21.

Martin, Marcel. "Cutadapt removes adapter sequences from high-throughput sequencing reads." *EMBnet. journal* 17.1 (2011): 10-12.

Wu, Tianzhi et al. "clusterProfiler 4.0: A universal enrichment tool for interpreting omics data." *Innovation (Cambridge (Mass.))* vol. 2,3 100141. 1 Jul. 2021, doi:10.1016/j.xinn.2021.100141